

Summary of Suas 2023

Video: [\(2129\) University College London - UCL Artificial Intelligence and Robotics \(UCL AIR\) | SUAS 2023 - YouTube](#)

University College London (UCL Artificial Intelligence and Robotics)

This team used a SIYI ZR10 4MP gimbal camera and 10x optical zoom / 30x hybrid zoom. The frame coverage is 30×17 and it they used k-means clustering and they extract the largest color contour and overlay it on a pink background sine the color pink was not mentioned in the regulation and therefore it is ignored during the k-means clustering. The shape classification was using a yolov8 model it was trained on a synthetic dataset of 10000 images and it was on 50 epochs and 5 batches each time. Optical character recognition is used to recognize the alphanumeric letter or digit printed on the ground once it has been isolated from the background rotating the symbol 15 degrees removes any orientation ambiguity. The target's pixel offset is first measured from the optical center of the image frame to the center of the detected bounding box. Using pinhole camera perspective geometry, these pixel offsets are scaled by the ratio of current flight altitude to the camera's focal length. This transformation converts two-dimensional image coordinates directly into real-world meter offsets relative to the drone's position.

They used RFD900 radios it is a 900MHz high-performance radio modem providing up to 40km range. Secondly the herelink ecosystem operates on the 2.4GHz frequency band with up to 20km range serving a secondary MAVlink and about the onboard computer they used the Nvidia jetson series companion computer the jetson handles high

performance on hard tasks such as object detection and classification and localization as well as interfacing with the flight controller via MAVROS

The onboard Jetson processes data and communicates with the Pixhawk orange cube flight controller then data is passed to the RFD900 900 MHz radio and this radio streams MAVlink packets over the air to a paired ground RFD9000 module connected via USB to the ground control station.